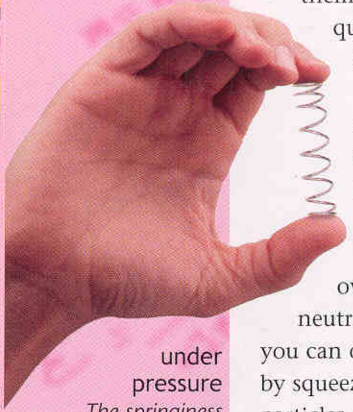


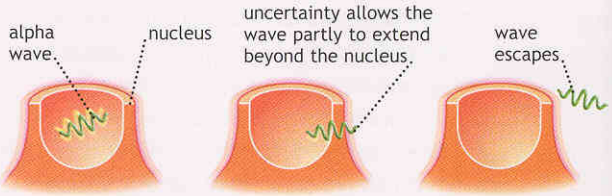
them, but they are able to "tunnel" through it. If quantum physics did not operate inside stars, the Sun would not shine and we would not be here.

The same tunneling process operates in reverse, enabling particles to get out of nuclei in the process of radioactive decay, or in nuclear fission. Particles in a nucleus are held together by a force called the strong nuclear force, which has a very short range (so nuclei are small) but overwhelms the electric repulsion of protons and neutrons once they are touching each other (just as you can overwhelm the expansionist tendency of a spring by squeezing it with your fingers). The effect is as if the particles are sitting in the deep crater of a volcano, which they would have to climb out of to escape, but once over the lip of the volcano, they would be rapidly repelled by the electric force. Tunneling enables some particles to escape from some nuclei without having enough energy to climb this barrier.

under pressure
The springiness of a spring can be overcome by a stronger squeeze, just as the electric repulsion between protons in a nucleus can be overcome by the strong force.

random decay

But there's another oddity about this process. In a collection of potentially radioactive nuclei, they do not all decay in this way at once. Individual nuclei decay at random while obeying the precise laws of chance. For a particular radioactive element, there is a characteristic



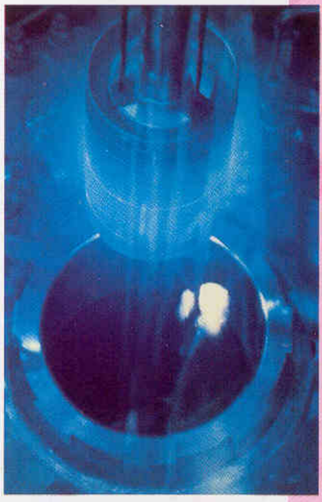
an alpha wave has insufficient energy to escape the strong force of the nucleus

there is a measurable probability that part of the wave extends outside the nucleus

the wave is then able to "tunnel" out and escape from the nucleus completely

tunneling out
Particles trapped inside the nucleus of an atom by the strong force can escape because of the waviness associated with quantum uncertainty.

time called the half-life. In one half-life, half the nuclei decay. In the next half-life, half the rest decay, and so on. So if you start with 128 nuclei with a half-life of 10 minutes, after 10 minutes, 64 have decayed, in the next 10 minutes, 32 decay, in the next 10 minutes, 16 decay, and so on (these are all average figures; in a real sample, the numbers decaying may differ slightly). But it is impossible to say in advance whether any particular nucleus will decay immediately, in 10 minutes' time, in half an hour, or whenever. Decay is as random as tossing a die, and Einstein hated the idea. So did Schrödinger, who picked up on this and combined it with the idea of the collapse of the wave function in his famous Cat Paradox (see page 32).



chain reaction
In a nuclear reactor, unstable nuclei of radioactive elements (such as uranium) absorb neutrons, making them "split" into two almost equal parts, releasing energy. As by-products of this fission, more neutrons are released, which then encourage more nuclei to split. A balance is maintained when each nucleus that splits triggers the splitting of just one more nucleus—thus creating a sustained chain reaction.

the hydrogen bond

Quantum physics also operates at the heart of life. The way atoms combine to form molecules is fully explained by quantum physics, which describes how electrons are shared between atoms to achieve the most stable combinations. The links between atoms that form molecules are called bonds, and they involve only the outermost electrons in the cloud around a nucleus. But there is one special case. A hydrogen atom consists of a single proton and a single electron, which, thanks to its quantum mechanical nature, still manages to surround the proton. There are no inner layers of electrons to conceal the proton when its lone electron is being shared with another nucleus in a conventional chemical bond.